

Repo: github.com/varyaaggarwal/fno-derivatives-analytics-platform | **Live app:** fnoderivativesanalyticsplatform.vercel.app | **API:** fando-derivatives-analytics-platform.onrender.com/api/health

Objective. A full-stack platform that pulls NSE/Upstox derivatives data, prices options and Greeks via Black-Scholes, solves implied volatility in reverse, renders the volatility surface, decomposes position P&L by Greek, explains the option chain in plain language, and runs a live signal engine + historical backtester for the DOS (Direction-of-SuperTrend) strategy on Bank Nifty weekly-expiry options.

Tech Stack

Layer	Technology	Role
Frontend	Next.js 14 (React 18) + TypeScript, Tailwind CSS, Recharts, Plotly.js	Option chain UI, P&L/interpretation cards, 3D vol surface, DOS panel
Backend	FastAPI (Python), Uvicorn	REST API – Greeks engine, IV solver, P&L decomposer, DOS engine
Live data	Upstox API v2 (REST) + Market Data Feed V3 (WebSocket, protobuf), NSE scrape fallback, yfinance	Option chain, spot price, real-time ticks
Database	Supabase (PostgreSQL)	Chain snapshot cache, DOS trade log
Math	NumPy, pandas, SciPy (Brent's method)	BSM pricing, IV solving, Greeks, SuperTrend
Deploy	Vercel (frontend), Render (backend, render.yaml IaC)	Hosting
Testing	pytest	30 tests across 6 files

Requirement Coverage

Requirement	Status	Where
Live option chain: OI, LTP, IV per strike	Done	/api/chain, upstox_chain.py
Greeks (Delta/Gamma/Theta/Vega) via BSM	Done	black_scholes.py
IV solver (reverse BSM, Brent's method)	Done	iv_solver.py
Volatility surface (3D, strike × expiry)	Done	/api/vol-surface, VolSurface3D.tsx
P&L decomposer by Greek	Done, interactive	pnl_decomposer.py, /pnl (Index/IV/Time sliders)
Plain-language cards (PCR, max pain, IV spike)	Done	interpretation.py
DOS: live SuperTrend signal + strike selector	Done	dos_strategy.py, /api/dos/live-signal
DOS: SL monitor (initial + trailing)	Done	/api/dos/sl-status
DOS: backtest ≥4 weeks, equity curve	Done (8 weeks default)	backtester.py, /api/dos/backtest
DOS trade log persisted (Supabase)	Done	supabase_client.py, persist=true
DOS interpretation card per trade	Done	interpretation.py
Deployed, working full stack	Done	Vercel + Render, see links above

Engineering Beyond MVP

- **Real-time WebSocket feed** (Upstox Market Data Feed V3, protobuf-decoded) as an opt-in push-based alternative to REST polling – `WS_FEED_ENABLED` , auth-failure "parking" so a bad token backs off cleanly instead of retry-storming.
- **Data provenance** – every chain response carries a structured `source_meta` object (which rung served it, capture time, staleness), not just a flat mock/live flag.
- **Holiday-aware market hours** – background poller checks NSE's real trading-holiday calendar (Upstox Holidays API), not just weekday+clock.
- **Render IaC** (render.yaml) and a frontend keepalive hook to avoid free-tier cold starts during a live demo.
- 30 automated tests (pytest) covering pricing, IV round-trips, holiday edge cases, provenance staleness, and auth-parking state transitions.

Key Design Decisions & Honest Caveats

- **Wed/Thu DOS cadence** matches the assignment spec exactly, but no longer matches live NSE mechanics as of July 2026; Bank Nifty weekly options were discontinued (Nov 2024); BNF now has only a monthly expiry. Implemented as specified, flagged here for the viva.
- **IV is solved, not trusted from the feed** – every row's last_price is run back through Brent's method (iv_solver.py); the feed's own IV field is only a fallback.
- **DOS backtest runs on statistically realistic mock 5-min candles** (correct hours/vol), since NSE has no free intraday BNF futures history; swappable for a paid vendor feed without touching backtester.py.
- **Backtest option premiums** priced via BSM at a flat assumed IV (14%), stated explicitly rather than hidden.
- **Assumptions:** same-day (ODTE) expiry both sessions, BNF lot size 15, risk-free rate 6.5%.

Validation Performed

- Put-call parity holds to 1e-8; matches Hull's textbook BSM example ($S=42, K=40, r=10\%, \sigma=20\%, T=0.5y \rightarrow C \approx 4.76$).
- IV solver round-trips price $\rightarrow$ solved σ to 1e-4; deep ITM/OTM delta sanity ($-1 / -0$).
- Excel workbook (reports/FnO_Analytics_Report.xlsx) recalculates with zero formula errors.